AIF-C01: Evaluating ML and Foundation Models (Exam Notes)

Golden rule: *first ask *what kind of output?* Category (classification) → confusion-matrix metrics. Number (regression) → error metrics. Generated text → ROUGE / BLEU / BERTScore / perplexity. Whole FM → benchmarks + human evaluation + Bedrock / Clarify.*

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1. Classification Metrics

1.1 Confusion Matrix

	Predicted Positive	***Predicted Negative***
Actual Positive	*TP (correct alarm)*	*FN (missed positive)*
Actual Negative	*FP (false alarm)*	*TN (correct rejection)*

- ***Definition:*** *a table comparing predictions with actual labels. All classification metrics are built from these 4 numbers.*
- ***Memory trick:*** *2nd word = what the model *predicted*, 1st word = was it *right*. "False Positive" = predicted positive, wrongly.*
- ***Running example (100 emails, 50 spam):*** *TP = 40, FN = 10, FP = 5, TN = 45.*
- ***Exam trap:*** *FP vs FN.* ***FP = innocent flagged (Type I error). FN = guilty missed (Type II error).***

1.2 Accuracy

- ***Definition:*** *share of all predictions that were correct.*
- ***Formula:*** *(TP + TN) / (TP + TN + FP + FN)*
- ***Use when:*** *classes are balanced and all errors cost the same.*
- ***Example:*** *(40 + 45) / 100 =* ***0.85***
- ***Exam trap (vs Recall/F1):*** *with imbalanced data, accuracy lies. 1,000 transactions with 10 frauds; a model that says "no fraud" always gets* ***99% accuracy but 0% recall****.* ***If the question says "imbalanced", never pick accuracy.***

1.3 Precision

- ***Definition:*** *of everything predicted positive, how much was truly positive.*
- ***Formula:*** *TP / (TP + FP)*
- ***Use when:*** *false positives are costly (spam filter, marking good customers as fraud).*
- ***Example:*** *40 / (40 + 5) =* ***0.889***
- ***Exam trap (vs Recall): Precision denominator = predicted positives (TP+FP). Recall denominator = actual positives (TP+FN).*** *Precision asks "can I trust my alarms?"*

1.4 Recall (Sensitivity, True Positive Rate)

- ***Definition:*** *of all actual positives, how many the model found.*
- ***Formula:*** *TP / (TP + FN)*
- ***Use when:*** *false negatives are costly (cancer screening, fraud detection).*
- ***Example:*** *40 / (40 + 10) =* ***0.80***
- ***Exam trap (vs Specificity): Recall = catches positives. Specificity = catches negatives.*** *Recall = Sensitivity = TPR (three names, one metric).*

1.5 Specificity (True Negative Rate)

- **Definition:** *of all actual negatives, how many the model correctly cleared.*
- **Formula:** *TN / (TN + FP)*
- **Use when:** *you must avoid false alarms among healthy/legit cases.*
- **Example:** *45 / (45 + 5) =* ***0.90***
- ***Exam trap (vs Precision): Specificity looks at actual negatives (TN+FP); Precision looks at predicted positives (TP+FP).*** *Also FPR = 1 – Specificity.*

1.6 F1 Score

- **Definition:** *harmonic mean of Precision and Recall.*
- **Formula:** *2 × (P × R) / (P + R)*
- **Use when:** *you need one number balancing FP and FN, especially on imbalanced data.*
- **Example:** *2 × (0.889 × 0.80) / (0.889 + 0.80) =* ***0.842***
- ***Exam trap (vs Accuracy): F1 ignores TN completely, so it is not fooled by a huge "negative" class.*** *Harmonic mean punishes a big gap: P = 1.0, R = 0.0 gives F1 = 0, not 0.5.*

1.7 AUC-ROC

- **Definition:** *ROC curve plots* ***TPR (recall) vs FPR*** *at every threshold; AUC is the area under it.*
- **Scale:** *0.5 = random guessing, 1.0 = perfect, below 0.5 = worse than random.*
- **Use when:** *comparing models across all thresholds; threshold-independent ranking quality.*
- **Example:** *Model A AUC = 0.92, Model B AUC = 0.78, so A separates classes better.*
- ***Exam trap (vs Accuracy/F1): Accuracy and F1 need one chosen threshold; AUC-ROC does not.*** *AUC-ROC can look optimistic on very imbalanced data (precision-recall curve is safer).*

1.8 Which Metric When

Situation	Costly error	Pick
Balanced data, equal costs	*Either*	*Accuracy*
Imbalanced data	*Depends*	*F1 / Recall / Precision (**not** Accuracy)*
Fraud detection	*Missing fraud (FN)*	***Recall***
Medical diagnosis	*Missing disease (FN)*	***Recall***
Spam filtering	*Real mail lost (FP)*	***Precision***
Need balance of P and R	*Both*	*F1*
Compare models over thresholds	*n/a*	*AUC-ROC*

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2. Regression Metrics

Running example: *actual = [10, 20, 30], predicted = [12, 18, 33] → errors = 2, –2, 3.*

2.1 MAE (Mean Absolute Error)

- **Definition:** *average size of errors, ignoring direction.*
- **Formula:** *(1/n) Σ |actual – predicted|*
- **Interpretation:** *same unit as target; "on average we're off by X". Robust to outliers.*

- **Example:** *(2 + 2 + 3) / 3 =* ***2.33***
- **Exam trap (vs RMSE): MAE treats all errors equally; RMSE punishes big errors more.** *RMSE ≥ MAE always.*

2.2 MSE (Mean Squared Error)

- **Definition:** *average of squared errors.*
- **Formula:** *(1/n) Σ (actual – predicted)²*
- **Interpretation:** *heavily penalizes large errors; units are squared (hard to read).*
- **Example:** *(4 + 4 + 9) / 3 =* ***5.67***
- **Exam trap (vs MAE): Squared units (dollars², kg²)**, *so it is not directly interpretable.*

2.3 RMSE (Root Mean Squared Error)

- **Definition:** *square root of MSE.*
- **Formula:** *√MSE*
- **Interpretation:** *back in the target's unit, still penalizes large errors.*
- **Example:** *√5.67 =* ***2.38***
- **Exam trap (vs MSE): RMSE = MSE with a square root, so the unit matches the target.**

2.4 R-squared (R^2, Coefficient of Determination)

- **Definition:** *proportion of variance in the target explained by the model.*
- **Formula:** *1 – (SS_residual / SS_total)*
- **Interpretation:** *1 = perfect, 0 = no better than predicting the mean, negative = worse than the mean.*
- **Example:** *mean = 20, SS_total = 200, SS_res = 17, so R^2 = 1 – 17/200 =* ***0.915*** *(91.5% of variance explained).*
- **Exam trap (vs RMSE): R^2 is unitless (relative fit); MAE/RMSE are in target units (absolute error).** *R^2 = 0.9 does not mean "90% accurate".*

Metric	Unit	Outlier sensitivity	Lower/Higher better
MAE	Same as target	Low	Lower
MSE	Squared	High	Lower
RMSE	Same as target	High	Lower
R^2	Unitless	Moderate	Higher

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3. Generative AI / Text Evaluation Metrics

3.1 ROUGE (Recall-Oriented Understudy for Gisting Evaluation)

- **Definition:** *overlap between generated text and reference text,* ***recall-focused****. Used for* ***summarization****.*
- **ROUGE-N:** *overlap of N-grams (ROUGE-1 = words, ROUGE-2 = word pairs).*
- **ROUGE-L:** *based on the* ***Longest Common Subsequence*** *(captures sentence-level word order).*
- **Example:** *Reference = "the cat sat on the mat"; Candidate = "the cat is on the mat".*
- *ROUGE-1 recall = 5/6 =* ***0.83***
- *ROUGE-2 recall = 3/5 =* ***0.60*** *(matches: the cat, on the, the mat)*
- *ROUGE-L: LCS = "the cat on the mat" (5 words) → 5/6 =* ***0.83***

- ***Exam trap (vs BLEU): ROUGE = recall ("did the summary capture the reference content?"), BLEU = precision ("is the translation made of correct reference words?").***

3.2 BLEU (Bilingual Evaluation Understudy)

- ***Definition:*** *N-gram* ***precision*** *of generated text vs reference(s), with a* ***brevity penalty*** *for too-short outputs. Used for* ***machine translation****.*
- ***Scale:*** *0 to 1 (often shown 0–100).*
- ***Example:*** *candidate is 4 words, reference 6 words → brevity penalty = e^(1 − 6/4) =* ***0.61****, which cuts the score even if all 4 words match.*
- ***Exam trap (vs ROUGE): Translation → BLEU. Summarization → ROUGE.*** *BLEU has a brevity penalty because precision alone rewards very short outputs.*

3.3 BERTScore

- ***Definition:*** *compares* ***contextual embeddings*** *(from BERT-like models) of generated vs reference tokens using cosine similarity. Captures* ***meaning****, not exact words.*
- ***Use when:*** *paraphrases, synonyms, or open-ended generation where wording varies.*
- ***Example:*** *"The car is fast" vs "The automobile is quick": ROUGE/BLEU low (few shared words), BERTScore ≈* ***0.92*** *(same meaning).*
- ***Exam trap (vs ROUGE/BLEU): ROUGE and BLEU count matching words; BERTScore compares meaning.*** *If the question says "semantic similarity" or "synonyms", pick BERTScore.*

3.4 Perplexity

- ***Definition:*** *how "surprised" a language model is by text; the exponential of the average negative log-likelihood.*
- ***Reading it: lower = better.*** *Perplexity 10 ≈ the model is as unsure as choosing among 10 equally likely next words.*
- ***Example:*** *model gives probability 0.25 to each correct next word → perplexity = 1 / 0.25 =* ***4****.*
- ***Exam trap (vs ROUGE/BLEU/BERTScore): Perplexity needs no reference text; it measures the language model's own fluency/prediction quality. The other three compare output to a reference.*** *It does not tell whether the answer is factually correct.*

3.5 Which Metric for Which Task

Task	Metric
Summarization	ROUGE
Machine translation	BLEU
Semantic similarity / paraphrase quality	BERTScore
Language model quality (no reference)	Perplexity
Classification, Q&A with fixed answers	Accuracy / F1

3.6 ROUGE vs BLEU vs BERTScore

Feature	ROUGE	BLEU	BERTScore
Main task	Summarization	Machine translation	Semantic similarity, generation
Focus	**Recall**	**Precision**	Meaning
Matches on	N-grams / LCS	N-grams	Embedding similarity
Handles synonyms	No	No	**Yes**
Brevity penalty	No	**Yes**	No
Needs reference	Yes	Yes	Yes
How to distinguish	"Summary covers the reference"	"Translation matches the reference"	"Same meaning, different words"

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4. Evaluating Foundation Models

4.1 Benchmark Datasets

Benchmark	What it tests	How to distinguish
GLUE	9 natural-language-understanding tasks (sentiment, entailment, similarity)	The original NLU suite
SuperGLUE	Harder NLU tasks	**Tougher successor to GLUE**
SQuAD	Reading comprehension / question answering from passages	**Q&A on Wikipedia passages**
MMLU	Knowledge across 57 subjects (math, law, medicine...)	**Multitask, exam-style knowledge**
HELM	Holistic evaluation: accuracy, robustness, fairness, bias, toxicity, efficiency	**Multi-metric, not just accuracy**
BIG-bench	200+ diverse, hard, unusual tasks	**Huge, broad, "beyond imitation game"**

- **Exam trap:** GLUE/SuperGLUE = language understanding; SQuAD = QA; MMLU = broad knowledge; **HELM = the only one focused on multiple trust dimensions (toxicity, bias).**

4.2 Automatic vs Human Evaluation

	Automatic	Human
Speed / cost	Fast, cheap, scalable	Slow, expensive
Consistency	High	Varies between raters
Best for	Accuracy, robustness, toxicity at scale	Style, tone, creativity, brand voice, helpfulness
Examples	ROUGE, BLEU, BERTScore, benchmarks	Rating scales, thumbs up/down, A/B comparison

- **How to distinguish: subjective qualities (tone, friendliness, brand voice) → human evaluation. Large-scale repeatable checks → automatic.**

4.3 AWS Tools

Amazon Bedrock Model Evaluation

- Compare and select FMs inside Bedrock.
- **Automatic evaluation:** built-in or your own datasets; tasks like text generation, summarization, Q&A, classification; metrics such as accuracy, robustness, toxicity.
- **Human evaluation:** your own workforce or an AWS-managed team; criteria like relevance, style, brand voice, helpfulness.
- Also supports a **model-as-a-judge** option (a model scores the outputs).

Amazon SageMaker Clarify (+ FMEval)

- Clarify detects **bias** in data and models and gives **explainability** (feature importance).
- **FMEval** is an open-source library (used with Clarify/SageMaker) to evaluate LLMs on **accuracy, toxicity, semantic robustness, prompt stereotyping, factual knowledge**.
- **How to distinguish: Bedrock Model Evaluation = pick/compare FMs in Bedrock (console, human + automatic). Clarify/FMEval = bias, explainability and programmatic FM evaluation in SageMaker.**

4.4 What to Evaluate

Dimension	Meaning	Example check
Accuracy	Correct, relevant outputs	ROUGE, F1, BERTScore
Toxicity	Harmful, offensive language	Toxicity score on test prompts
Robustness	Stable when input is slightly changed (typos, paraphrases)	Compare outputs on perturbed prompts
Bias / fairness	Equal treatment across groups	Compare results across demographic groups, stereotype tests

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5. Business Metrics for AI Solutions

Metric	Formula / Meaning	Example	Exam trap
ROI	(Benefit – Cost) / Cost × 100	Cost $50k, benefit $80k → **60%**	**ROI = return relative to cost**; not the same as raw profit
Cost per user	Total AI cost / number of users	$10,000 / 2,000 users = **$5**	Measures efficiency of spend, not value delivered
Customer satisfaction (CSAT)	Satisfied responses / total responses × 100	850 / 1,000 = **85%**	Measures experience, not revenue
Conversion rate	Conversions / visitors × 100	50 / 1,000 = **5%**	Revenue-side metric (recommendations, chatbots for sales)
Efficiency gains	Time or cost saved vs before	Ticket handling 10 min → 6 min = **40%** saved	Measures productivity, not quality

- **How to distinguish: model metrics (F1, ROUGE) say if the model is good; business metrics say if the solution is worth it.** A model with great F1 can still have poor ROI.

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6. Model Performance Problems

	Underfitting	Good fit	Overfitting
Training error	**High**	Low	**Very low**
Test/validation error	**High**	Low	**High**
Bias / Variance	**High bias**, low variance	Balanced	Low bias, **high variance**
Cause	Model too simple, too few features/training	Right complexity	Model too complex, memorizes noise
Fix	More complex model, more features, train longer	n/a	More data, regularization, simpler model, early stopping, dropout

- **Bias:** error from wrong/simple assumptions (misses patterns). **Variance:** error from being too sensitive to training data (learns noise).
- **Tradeoff:** lowering bias (more complex model) raises variance, and vice versa. Goal: the sweet spot with lowest total test error.
- **Example:** Train accuracy 99%, test accuracy 70% → **overfitting**. Train 60%, test 58% → **underfitting**.
- **How metrics reveal it: compare training vs test metrics. Big gap → overfitting. Both poor → underfitting.**
- **Exam trap:** overfitting is *not* "low training accuracy"; **it is high training performance with poor unseen-data performance.**

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7. Master Comparison Table

Metric	Task type	Measures	Use when
Accuracy	Classification	Overall correctness	Balanced classes
Precision	Classification	Correctness of positive predictions	FP costly (spam)
Recall	Classification	Coverage of actual positives	FN costly (fraud, disease)
Specificity	Classification	Coverage of actual negatives	Avoid false alarms
F1	Classification	Balance of P and R	Imbalanced data
AUC-ROC	Classification	Class separation across thresholds	Compare models
MAE	Regression	Average absolute error	Outliers should not dominate
MSE	Regression	Average squared error	Penalize large errors
RMSE	Regression	Squared error in target units	Penalize large errors, readable
R^2	Regression	Variance explained	Overall fit
ROUGE	Text generation	Recall of reference n-grams / LCS	Summarization
BLEU	Text generation	N-gram precision + brevity penalty	Translation
BERTScore	Text generation	Embedding similarity	Meaning, paraphrase
Perplexity	Language modeling	Model's prediction uncertainty	LM quality, no reference
ROI	Business	Return vs cost	Justify AI investment
CSAT	Business	User happiness	Chatbot/assistant experience

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8. Scenario → Correct Metric (20)

#	Scenario	Metric
1	Detect credit card fraud (0.1% fraud)	**Recall** (or F1)
2	Diagnose a serious disease, missing a case is dangerous	**Recall**
3	Spam filter must not send real mail to spam	**Precision**
4	Balanced cat/dog image classifier	Accuracy
5	Need a single score balancing P and R on imbalanced data	**F1**
6	Compare two classifiers across all thresholds	**AUC-ROC**
7	Of healthy patients, how many correctly cleared	**Specificity**
8	Predict house prices, average error in rupees	**MAE**
9	Predict demand, large misses are very costly	**RMSE**
10	How much variance in sales does the model explain	**R^2**
11	Evaluate a news article summarizer vs human summaries	**ROUGE**
12	Evaluate English → Hindi translation	**BLEU**
13	Judge paraphrase quality where wording differs	**BERTScore**
14	Measure how well an LLM predicts the next word, no reference	**Perplexity**
15	Test LLM broad knowledge across 57 subjects	**MMLU**
16	Evaluate an LLM on accuracy, bias, toxicity and robustness together	**HELM**
17	Rate chatbot tone and brand voice	**Human evaluation** (Bedrock)
18	Detect bias in training data and explain predictions	**SageMaker Clarify**
19	Does the chatbot investment pay off financially	**ROI**
20	Model: train 98%, test 65%	**Overfitting** (high variance)

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9. Precision vs Recall vs F1: The Mango Basket

A basket has **100 mangoes, 20 are ripe**. Your job: pick the ripe ones.

Strategy A (careful): pick 10, and 8 are ripe.

- Precision = 8 / 10 = **0.80** (most of what you picked is ripe)
- Recall = 8 / 20 = **0.40** (you found only 40% of ripe mangoes)
- F1 = 2 × 0.8 × 0.4 / 1.2 = **0.53**

Strategy B (greedy): pick 50, and 18 are ripe.

- Precision = 18 / 50 = **0.36** (lots of unripe ones picked)
- Recall = 18 / 20 = **0.90** (you found nearly all ripe ones)
- F1 = 2 × 0.36 × 0.9 / 1.26 = **0.51**

	Question it answers	Denominator	Boosted by
Precision	"Of what I picked, how much is ripe?"	Picked (TP+FP)	Picking fewer, safer items
Recall	"Of all ripe ones, how many did I find?"	All ripe (TP+FN)	Picking more items
F1	"Am I decent at both?"	Combines both	Balance

- **How to distinguish: Precision = quality of your picks. Recall = completeness of your picks.** They pull in opposite directions; F1 punishes lopsided results.
- **Real-world mapping:** spam filter → precision. Cancer screening → recall.

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10. Exam-Style MCQs

Q1. A bank builds a fraud model; only 0.1% of transactions are fraud. Which metric is most appropriate? A) Accuracy B) Recall C) R^2 D) Perplexity **Answer: B.** Missing fraud (FN) is costly and accuracy is misleading on imbalanced data.

Q2. An email service wants to minimize legitimate emails being marked as spam. Which metric to maximize? A) Recall B) Precision C) MAE D) BLEU **Answer: B.** Precision reduces false positives.

Q3. Which metric is most commonly used to evaluate text summarization against reference summaries? A) BLEU B) ROUGE C) RMSE D) AUC **Answer: B.** ROUGE measures n-gram/LCS overlap (recall-oriented).

Q4. Which metric evaluates machine translation quality using n-gram precision and a brevity penalty? A) ROUGE B) BERTScore C) BLEU D) Perplexity **Answer: C.** BLEU.

Q5. A team needs to judge whether a generated sentence has the same meaning as the reference even with different words. Which metric? A) BLEU B) ROUGE-N C) BERTScore D) MSE **Answer: C.** BERTScore compares embeddings (semantics).

Q6. A language model A has perplexity 12 and model B has perplexity 45 on the same data. Which is true? A) B is better B) A is better C) They are equal D) Perplexity cannot compare models **Answer: B.** Lower perplexity means better next-word prediction.

Q7. Which regression metric penalizes large errors more heavily while staying in the same unit as the target? A) MAE B) MSE C) RMSE D) R^2 **Answer: C.** RMSE squares errors (penalizes big ones), then square-roots back to the original unit.

Q8. A regression model has $R^2 = 0.92$. What does this mean? A) 92% of predictions are exactly right B) The model explains 92% of the variance in the target C) Average error is 0.92 D) The model is overfitted **Answer: B.** R^2 is the proportion of variance explained.

Q9. A model scores 99% on training data and 70% on test data. What is the problem, and a fix? A) Underfitting; add features B) Overfitting; add regularization or more data C) High bias; train longer D) Data leakage only **Answer: B.** Large train/test gap = overfitting (high variance).

Q10. Which benchmark measures LLM knowledge across 57 subjects such as law, math and medicine? A) SQuAD B) GLUE C) MMLU D) HELM **Answer: C.** MMLU.

Q11. Which framework evaluates language models across multiple dimensions including accuracy, robustness, fairness, bias and toxicity? A) GLUE B) SQuAD C) HELM D) ROUGE **Answer: C.** HELM is holistic and multi-metric.

Q12. A company wants to compare FMs on brand voice and friendliness using its own reviewers inside AWS. Which option fits? A) Bedrock Model Evaluation with human evaluation B) ROUGE score C) SageMaker Data Wrangler D) AUC-ROC **Answer: A.** Subjective qualities need human evaluation, supported by Bedrock Model Evaluation.

Q13. A binary classifier has AUC-ROC of 0.5. What does this indicate? A) Perfect model B) Random guessing C) 50% precision D) Underfitting only **Answer: B.** 0.5 means no better than chance.

Q14. *A firm spent $50,000 on a chatbot and gained $80,000 in benefit. Which business metric gives 60%? A) Conversion rate B) CSAT C) ROI D) Cost per user* ***Answer: C.*** *ROI = (80k – 50k) / 50k = 60%.*

Q15. *Why is F1 preferred over the simple average of precision and recall? A) It ignores false negatives B) The harmonic mean penalizes a big gap between precision and recall C) It needs no threshold D) It is used for regression* ***Answer: B.*** *Harmonic mean drops sharply if either value is low.*

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11. Last-Minute Revision Sheet (20 Points)

1. ***Precision = TP/(TP+FP); Recall = TP/(TP+FN); Specificity = TN/(TN+FP).***
2. *Recall = Sensitivity = TPR (same metric).*
3. ***Imbalanced data → avoid Accuracy; use Recall, Precision or F1.***
4. *Fraud and disease detection →* ***Recall*** *(FN costly). Spam filter →* ***Precision*** *(FP costly).*
5. ***F1 = harmonic mean of P and R; ignores TN.***
6. ***AUC-ROC: TPR vs FPR; 0.5 = random, 1.0 = perfect; threshold-independent.***
7. *FP = Type I error (false alarm); FN = Type II error (missed case).*
8. ***MAE = average absolute error (same unit, outlier-robust).***
9. ***MSE = squared units; RMSE = √MSE (same unit, penalizes big errors).*** *RMSE ≥ MAE.*
10. ***R^2 = variance explained; unitless; 1 best; can be negative.***
11. ***ROUGE → summarization, recall-based (ROUGE-N = n-grams, ROUGE-L = longest common subsequence).***
12. ***BLEU → translation, precision-based with brevity penalty.***
13. ***BERTScore → semantic similarity via embeddings; handles synonyms.***
14. ***Perplexity → LM quality, lower is better, needs no reference text.***
15. *Benchmarks:* ***GLUE/SuperGLUE = NLU, SQuAD = reading QA, MMLU = 57-subject knowledge, HELM = holistic (accuracy, bias, toxicity, robustness), BIG-bench = 200+ diverse tasks.***
16. ***Subjective quality (tone, brand voice) → human evaluation; scale and repeatability → automatic.***
17. ***Bedrock Model Evaluation = compare FMs (automatic, human, model-as-judge). SageMaker Clarify/FMEval = bias, explainability, accuracy/toxicity/robustness checks.***
18. *FM evaluation dimensions:* ***accuracy, toxicity, robustness, bias/fairness.***
19. *Business metrics:* ***ROI = (benefit – cost)/cost; also cost per user, CSAT, conversion rate, efficiency gains.*** *Good model metrics do not guarantee good ROI.*
20. ***Overfitting = low train error, high test error (high variance). Underfitting = high error on both (high bias). Bias-variance tradeoff: reducing one tends to raise the other.***